

Name : _____

Teacher/ Class : _____

SYDNEY TECHNICAL HIGH SCHOOL



HSC ASSESSMENT TASK 1

DECEMBER 2009

MATHEMATICS - EXTENSION 1

Time Allowed : **70 minutes**

Instructions:

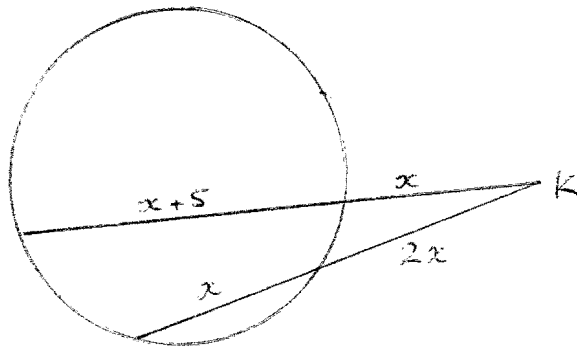
- Write your name and class at the top of each page.
- All necessary working must be shown. Marks may be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Start **each** question on a new **page**.
- Diagrams, unless otherwise, stated are not to scale.

Q1	Q2	Q3	Q4	Q5	TOTAL
/10	/10	/10	/10	/10	/50

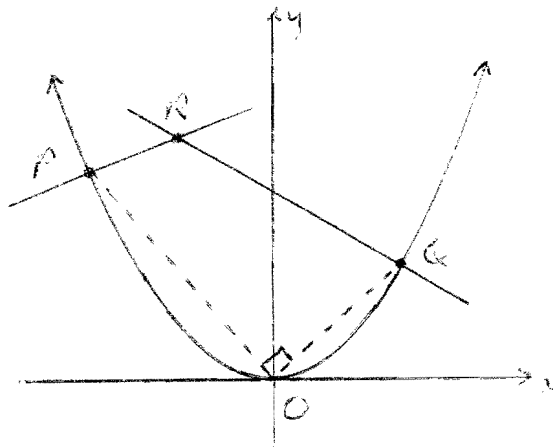
Question 2

- a) Secants to the circle are drawn to meet at K . Find the value of x .

2



- b)



For the parabola $x = 2at, y = at^2$,
points P and Q subtend a right angle
at the vertex.

The normals at P and Q intersect at R .

- Prove that $pq = -4$, where p and q are the parameters for P and Q . 1
- Derive the gradient of the normal at P and show that its equation is $yp - ap^3 = -x + 2ap$. 2
- R has a y coordinate of $ap^2 - 2a + aq^2$. Find its x coordinate. 2
- Show that the cartesian locus for R is the parabola $x^2 = 16a(y - 6a)$. 2
- Find the coordinates of the focus of the parabola in iv). 1

Question 3

- a) Accurately describe how to find the centre of the circle passing through points A, B, C .

1

B.

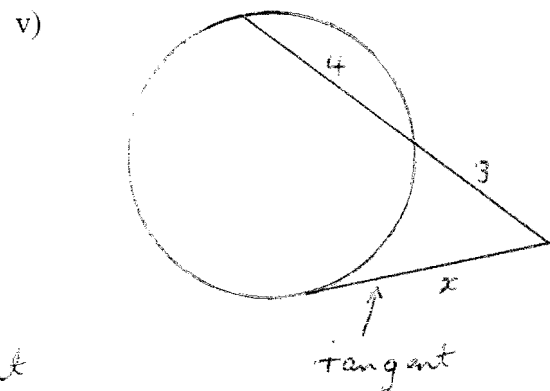
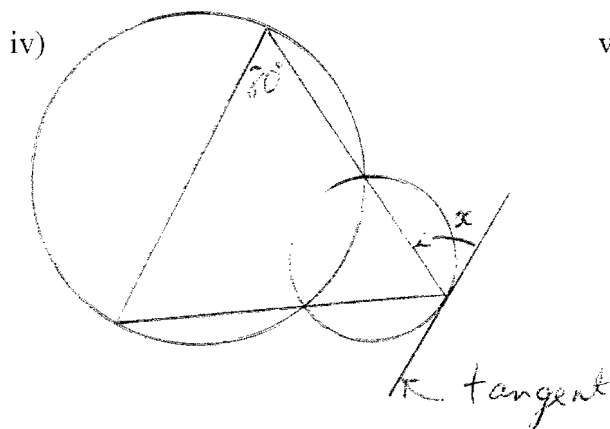
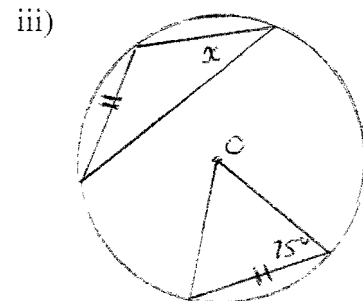
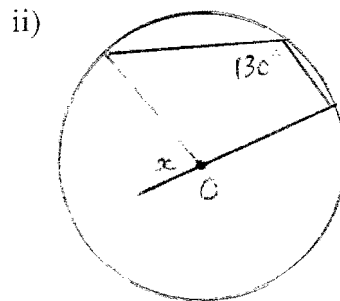
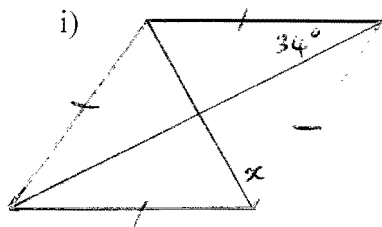
C.

A.

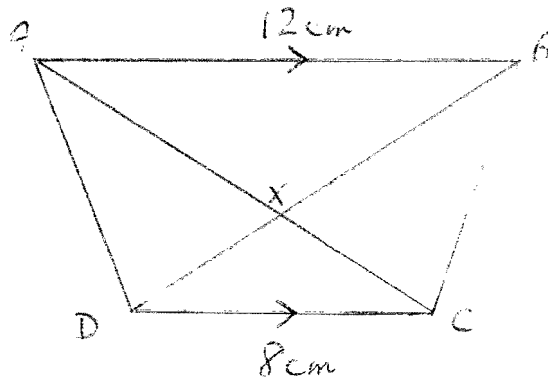
Question 1

- a) Find the value of each pronumeral. O is the centre of each circle. Reasons are not necessary.

5



- b) $ABCD$ is a trapezium with $AB \parallel DC$. $AB = 12\text{cm}$, $DC = 8\text{cm}$, $AC = 9\text{cm}$. Diagonals intersect at X .



- i) State which two triangles are similar (do not prove similarity). 1
- ii) Hence, find the length of AX 2
- c) Find the first negative term of the sequence 496, 489, 482, 2

Question 5

- a) The sum of the first two terms of a geometric series is 6, and the sum of the second and third terms is -5. 1

Find the common ratio for this series.

- b) Kenny begins his retirement with \$500,000 and invests it to earn 6% p.a. Interest is calculated on the balance at the end of each month and added to the remaining balance.

Immediately after the interest calculation, Kenny withdraws an amount, M , for living expenses.

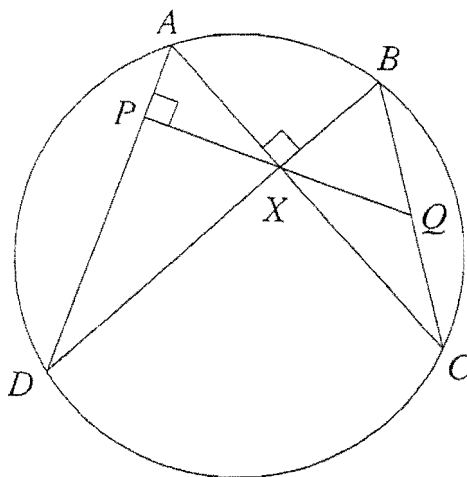
- i) Let A_n be the balance remaining after the n th withdrawal.

Show that $A_2 = 500000 \times 1.005^2 - 1.005M - M$ 1

- ii) How much can he afford to withdraw each month if he plans to live for another 20 years? 2

- iii) If, instead, he withdraws \$5000 per month, show that the number of months can be found using $1.005^n = 2$. (do not solve this). 1

c)



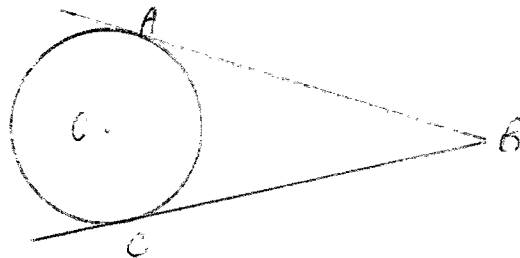
NOT
TO
SCALE

The diagram shows points, A, B, C and D on a circle. The lines AC and BD are perpendicular and intersect at X . The perpendicular to AD through X meets AD at P and BC at Q .

- i) Neatly redraw the diagram onto your answer page. 3
- ii) Prove that $\angle QXB = \angle QBX$. 3
- iii) Prove that Q bisects BC . 2

END OF PAPER

- b) Tangents BA and BC are drawn to the circle, with centre O .



- i) Neatly redraw the above diagram onto your answer page.
 - ii) Prove that $AB = CB$ 2
 - iii) Join AC and OB . Prove that $AC \perp OB$ 3
- c) Express $0.6\dot{5}$ as the sum of an infinite geometric series and find its limiting sum in fraction form. 2
- d) For a particular series, the sum of the first n terms is given by $S_n = 3n^2 - 2n$. Find the simplified expression for the $(n + 1)$ th term, T_{n+1} 2

Question 4

- a) The first three terms of a certain geometric series are: $x + 2 + 1\frac{1}{2} + \dots$
- i) Find the value of x 1
 - ii) Find an expression for the tenth term in the form $\frac{3^a}{2^b}$ 1
- b) Evaluate $\sum_{n=10}^{30} (2^n + 2n + 2)$ 3
- c) John invests \$1000 into a savings account.
- i) If the account earns 6% p.a. compounded annually, find the account's value at the end of 15 years. 1
 - ii) If the account is to have a value of \$5000 after 15 years, find the annual compound interest rate needed to achieve this. Give your answer correct to 1 decimal place. 2
- d) Mary invests \$1000 into an account, earning 6% p.a. interest, at the beginning of 2010. 2
She continues to deposit \$1000 at the beginning of each subsequent year into the account.
Find the total value of her savings at the end of 2024.

SOLUTIONS

① a) i) 56° ii) 80° iii) 15°

iv) 80° v) $x = \sqrt{21}$

b) i) $\triangle ABX$ & $\triangle CDX$

ii) $\frac{AX}{XC} = \frac{3}{2}$

$2AX = 3XC$

$2AX = 3(9 - AX)$

$5AX = 27$

$\therefore \underline{AX = 5.4 \text{ cm}}$

c) Need $T_n = 496 + (n-1)(-7) < 0$

$\therefore 496 - 7n + 7 < 0$

$-7n < -503$

$n > \frac{503}{7}$

$\therefore \underline{T_{72}(-1)}$ is first neg. term

② a) $x(2x+5) = 2x(3x)$

$2x^2 + 5x = 6x^2$

$4x^2 - 5x = 0$

$x(4x-5) = 0$

$x = 0$ or $\frac{5}{4}$ ($x > 0$)

$\therefore \underline{x = \frac{5}{4}}$

b) i) $M_{OP} \times M_{OQ} = -1$

$\therefore \frac{ap^2}{2ap} \times \frac{aq^2}{2aq} = -1$

$\therefore \frac{p}{2} \times \frac{q}{2} = -1$

$\therefore \underline{pq = -4}$

ii) $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} \rightarrow \text{at } P, M_T = p$
 $= 2at \times \frac{1}{2a} \therefore m_N = \underline{\underline{-\frac{1}{p}}}$
 $= t$

eqn. of normal at P is

$y - ap^2 = -\frac{1}{p}(x - 2ap)$

$\therefore \underline{yp - ap^3 = -x + 2ap} \quad (*)$

(Similarly for Q: $yq - aq^3 = -x + 2aq$)

iii) Sub $y = ap^2 - 2a + aq^2$ into $(*)$ above

$\therefore (ap^2 - 2a + aq^2)p - ap^3 = -x + 2ap$

$\therefore x = ap^3 - ap^2 + 2ap - aq^2p + 2ap$

$= 4ap - aq^2p$

$= \underline{\underline{4ap + 4aq}}$

iv) $x = 4ap + 4aq = 4a(p+q)$

$\therefore p+q = \frac{x}{4a}$

and $y = a(p^2 + q^2 - 2)$

$= a[(p+q)^2 - 2pq - 2]$

$= a\left[\left(\frac{x}{4a}\right)^2 - 2(-4) - 2\right]$

$= a\left(\frac{x^2}{16a^2} + 8 - 2\right)$

$= \frac{x^2}{16a} + 6a$

$\therefore y - 6a = \frac{x^2}{16a}$

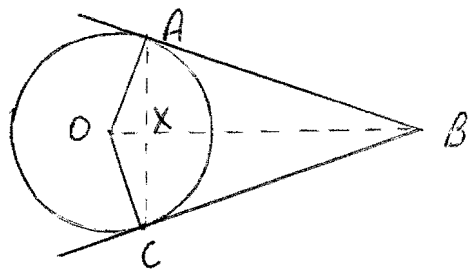
$\therefore x^2 = 16a(y - 6a)$ as reqd.

v) Focal length $4a$, vertex $(0, 6a)$

$\therefore \underline{\underline{\text{focus at } (0, 10a)}}$

③ a) Centre is at point of intersection of two perpendicular bisectors of say, AB and AC.

b) i)



ii) $OA = OB$ (equal radii)
 OB is common
 $\angle OAB = \angle OCB = 90^\circ$
 (radius $\perp$ tangent)

$\therefore \triangle OAB \equiv \triangle OCB$ (RHS)

$\therefore AB = BC$ (corresponding sides in congruent triangles)

iii) $\angle AOX = \angle COX$ (corresponding angles in congruent triangles)

OX is common

$OA = OC$ (above)

$\therefore \triangle AOX \equiv \triangle COX$ (SAS)

$\therefore \angle AXO = \angle CXO$ (corresponding angles in congruent triangles)

$\therefore \angle AXO \text{ \& } \angle CXO = 90^\circ$ each.
 (AXC is a straight line)

$\therefore AC \perp OB$ as reqd.

$$\begin{aligned} \text{c) } 0.6\bar{5} &= \frac{6}{10} + \frac{5}{100} + \frac{5}{1000} + \dots \\ &= \frac{6}{10} + \frac{a}{1-r} \\ &= \frac{6}{10} + \frac{\frac{5}{100}}{1-\frac{1}{10}} \\ &= \frac{6}{10} + \frac{5}{100} \times \frac{10}{9} \\ &= \frac{6}{10} + \frac{5}{90} \\ &= \frac{59}{90} \end{aligned}$$

$$\begin{aligned} \text{d) } T_{n+1} &= S_{n+1} - S_n \\ &= 3(n+1)^2 - 2(n+1) - 3n^2 + 2n \\ &= 3n^2 + 6n + 3 - 2n - 2 - 3n^2 + 2n \\ &= \underline{\underline{6n+1}} \end{aligned}$$

$$\begin{aligned} \text{④ a) i) } \frac{2}{x} &= \frac{1\frac{1}{2}}{2} = \frac{3}{4} \\ 3x &= 8 \\ \underline{\underline{x}} &= \underline{\underline{\frac{8}{3}}} \end{aligned}$$

$$\begin{aligned} \text{ii) } T_{10} &= ar^9 \\ &= \frac{8}{3} \times \left(\frac{3}{4}\right)^9 \\ &= \frac{2^3}{3} \times \frac{3^9}{2^{18}} \\ &= \underline{\underline{\frac{3^8}{2^{15}}}} \end{aligned}$$

$$b) \sum_{n=10}^{30} (2^n) + \sum_{n=10}^{30} (2n+2)$$

$$= \frac{2^{10}(2^{21}-1)}{2-1} + \frac{21}{2}(22+62)$$

$$= \underline{\underline{2,147,483,506}}$$

$$c) i) A = 1000(1.06)^{15}$$

$$= \underline{\underline{\$2396.56}}$$

$$ii) 5000 = 1000(1+r)^{15}$$

$$(1+r)^{15} = 5$$

$$r = \sqrt[15]{5} - 1$$

$$= 0.1132 \dots$$

$$\underline{\underline{\div 11.3\% \text{ p.a.}}}$$

$$d) \text{ First } \$1000 \Rightarrow 1000(1.06)^{15}$$

$$\text{Second } \$1000 \Rightarrow 1000(1.06)^{14}$$

$$\vdots$$

$$\text{Last } \$1000 \Rightarrow 1000(1.06)^1$$

$$\therefore \text{ total} = \frac{1000(1.06)(1.06^{15}-1)}{1.06-1}$$

$$= \underline{\underline{\$24672.53}}$$

$$(5) a) a + ar = a(1+r) = 6$$

$$ar + ar^2 = ar(1+r) = -5$$

$$\therefore r = \underline{\underline{-\frac{5}{6}}}$$

$$b) i) A_1 = 500000 \times 1.005 - M$$

$$A_2 = A_1 \times 1.005 - M$$

$$= (500000 \times 1.005 - M) \times 1.005 - M$$

$$= 500000 \times 1.005^2 - 1.005M - M$$

$$ii) A_{240} = 500000 \times 1.005^{240} - (1.005^{239}M$$

$$- \dots - M$$

$$= 500000 \times 1.005^{240} - (1.005^{239}M$$

$$+ \dots + M)$$

$$= 500000 \times 1.005^{240} - M \frac{(1.005^{240} - 1)}{1.005 - 1}$$

$$\text{and } A_{240} = 0,$$

$$\therefore M = \frac{(500000 \times 1.005^{240}) \times 0.005}{1.005^{240} - 1}$$

$$= \underline{\underline{\$3582.16 \text{ per month.}}}$$

$$iii) 5000 = \frac{(500000 \times 1.005^n) \times 0.005}{1.005^n - 1}$$

$$\therefore 5000 \times 1.005^n - 5000 = 2500 \times 1.005^n$$

$$\therefore 2500 \times 1.005^n = 5000$$

$$\therefore 1.005^n = 2$$

as req'd.